

Tutorial 6: Basic Statistical Analysis in R

Descriptive Statistics · Hypothesis Testing · Correlation · Statistical Report

2026-02-22

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About This Tutorial

In this tutorial we will perform a full statistical analysis on a real medical dataset, step by step.

By the end of this session you will be able to:

- Explore and summarize a dataset using descriptive statistics
- Run and interpret a t-test and a chi-square test
- Compute and visualize a correlation matrix
- Understand how this document *itself* is a statistical report

Dataset: `Pima.tr` from the `MASS` package — diabetes data from Pima Indian women.

Part 0 — Setup

Load Packages

```
library(MASS)      # contains the Pima.tr dataset
library(psych)     # describe() and describeBy()
library(ggpubr)    # ggscatter() for correlation plots
```

Load the Data

```
data(Pima.tr)
```

Variable Reference

Variable	Description	Type
<code>npreg</code>	Number of pregnancies	Continuous
<code>glu</code>	Plasma glucose concentration	Continuous
<code>bp</code>	Diastolic blood pressure (mm Hg)	Continuous
<code>skin</code>	Triceps skin fold thickness (mm)	Continuous
<code>bmi</code>	Body mass index	Continuous
<code>ped</code>	Diabetes pedigree function (family history score)	Continuous
<code>age</code>	Age (years)	Continuous

Variable	Description	Type
type	Diabetes diagnosis	Categorical (Yes / No)

Part 1 — First Look at the Data

Basic Exploration

```
# Dimensions: how many rows and columns?
dim(Pima.tr)
```

```
## [1] 200  8
```

```
# First 6 rows
head(Pima.tr)
```

```
##  npreg glu bp skin  bmi  ped age type
## 1     5  86 68  28 30.2 0.364 24  No
## 2     7 195 70  33 25.1 0.163 55  Yes
## 3     5  77 82  41 35.8 0.156 35  No
## 4     0 165 76  43 47.9 0.259 26  No
## 5     0 107 60  25 26.4 0.133 23  No
## 6     5  97 76  27 35.6 0.378 52  Yes
```

```
# Variable types and structure
str(Pima.tr)
```

```
## 'data.frame':  200 obs. of  8 variables:
##  $ npreg: int  5 7 5 0 0 5 3 1 3 2 ...
##  $ glu  : int  86 195 77 165 107 97 83 193 142 128 ...
##  $ bp   : int  68 70 82 76 60 76 58 50 80 78 ...
##  $ skin : int  28 33 41 43 25 27 31 16 15 37 ...
##  $ bmi  : num  30.2 25.1 35.8 47.9 26.4 35.6 34.3 25.9 32.4 43.3 ...
##  $ ped  : num  0.364 0.163 0.156 0.259 0.133 ...
##  $ age  : int  24 55 35 26 23 52 25 24 63 31 ...
##  $ type : Factor w/ 2 levels "No","Yes": 1 2 1 1 1 2 1 1 1 2 ...
```

How Many Diabetic vs. Non-Diabetic?

```
# Count per group
table(Pima.tr$type)
```

```
##
##  No Yes
## 132  68
```

```
# Proportions
prop.table(table(Pima.tr$type))
```

```
##
##   No   Yes
## 0.66 0.34
```

Missing Values

```
# Total missing values in the dataset
sum(is.na(Pima.tr))
```

```
## [1] 0
```

```
# Missing values per column
colSums(is.na(Pima.tr))
```

```
## npreg   glu    bp  skin   bmi   ped   age   type
##      0     0     0     0     0     0     0     0
```

Try it yourself: What type of variable is type? What type is glu? Is this dataset balanced between the two groups?

Part 2 — Descriptive Statistics

Summary of the Entire Dataset

```
summary(Pima.tr)
```

```
##      npreg      glu      bp      skin
##  Min.   : 0.00   Min.   : 56.0   Min.   : 38.00   Min.   : 7.00
## 1st Qu.: 1.00   1st Qu.:100.0   1st Qu.: 64.00   1st Qu.:20.75
## Median : 2.00   Median :120.5   Median : 70.00   Median :29.00
## Mean   : 3.57   Mean   :124.0   Mean   : 71.26   Mean   :29.21
## 3rd Qu.: 6.00   3rd Qu.:144.0   3rd Qu.: 78.00   3rd Qu.:36.00
## Max.   :14.00   Max.   :199.0   Max.   :110.00   Max.   :99.00
##      bmi      ped      age      type
##  Min.   :18.20   Min.   :0.0850   Min.   :21.00   No :132
## 1st Qu.:27.57   1st Qu.:0.2535   1st Qu.:23.00   Yes: 68
## Median :32.80   Median :0.3725   Median :28.00
## Mean   :32.31   Mean   :0.4608   Mean   :32.11
## 3rd Qu.:36.50   3rd Qu.:0.6160   3rd Qu.:39.25
## Max.   :47.90   Max.   :2.2880   Max.   :63.00
```

Detailed Statistics with psych

`psych::describe()` gives us more: mean, SD, median, skewness, kurtosis, and more — all in one table.

```
describe(Pima.tr[, -8]) # exclude the categorical variable 'type'
```

```
##      vars  n   mean    sd median trimmed   mad   min   max range skew
## npreg    1 200   3.57   3.37   2.00    3.11   2.97   0.00  14.00  14.0 1.09
## glu     2 200 123.97 31.67 120.50 122.08 31.88 56.00 199.00 143.0 0.45
## bp      3 200  71.26 11.48  70.00   71.13 11.86 38.00 110.00  72.0 0.16
## skin    4 200  29.22 11.72  29.00   28.80 11.86   7.00  99.00  92.0 1.08
## bmi     5 200  32.31   6.13  32.80   32.26   6.52 18.20  47.90  29.7 0.02
## ped     6 200   0.46   0.31   0.37   0.42   0.23   0.09   2.29   2.2 1.94
## age     7 200  32.11 10.98  28.00   30.46   8.90 21.00  63.00  42.0 1.08
##      kurtosis   se
## npreg      0.44 0.24
## glu      -0.38 2.24
## bp       0.51 0.81
## skin     4.88 0.83
## bmi     -0.44 0.43
## ped      6.49 0.02
## age      0.23 0.78
```

Key columns to focus on:

- mean / median — are they close? If not, the distribution may be skewed
- sd — how spread out are the values?
- skew — positive skew means a long right tail
- min / max — are there suspicious extreme values?

Descriptive Statistics by Group

Now let's split by diabetes status to compare the two groups:

```
describeBy(Pima.tr[, -8], group = Pima.tr$type)
```

```
##
## Descriptive statistics by group
## group: No
##      vars  n   mean    sd median trimmed   mad   min   max range skew
## npreg    1 132   2.92   2.81   2.00    2.47   1.48   0.00  13.0  13.00 1.53
## glu     2 132 113.11 26.64 109.50 111.46 21.50 56.00 193.0 137.00 0.62
## bp      3 132  69.55 11.08  70.00   69.37 10.38 38.00 110.0  72.00 0.25
## skin    4 132  27.20 10.93  27.00   26.73 13.34   8.00  60.0  52.00 0.39
## bmi     5 132  31.07   6.38  31.05   30.88   7.19 18.20  47.9  29.70 0.29
## ped     6 132   0.42   0.27   0.32   0.38   0.19   0.09   1.7   1.61 1.52
## age     7 132  29.23   9.54  26.00   27.46   5.93 21.00  63.0  42.00 1.70
##      kurtosis   se
## npreg      2.16 0.24
## glu       0.45 2.32
## bp       0.64 0.96
```

```
## skin      -0.54 0.95
## bmi       -0.40 0.56
## ped       3.24 0.02
## age       2.53 0.83
## -----
## group: Yes
##      vars  n   mean    sd median trimmed   mad   min    max   range  skew
## npreg    1 68   4.84   3.97   4.50    4.54   5.19   0.00   14.00   14.00   0.41
## glu     2 68 145.06 30.12 144.00  145.14  34.10  80.00  199.00  119.00   0.02
## bp      3 68  74.59 11.58  76.00   74.46  11.86  40.00  106.00   66.00  -0.03
## skin    4 68  33.12 12.30  32.00   32.71   8.90   7.00   99.00   92.00   2.02
## bmi     5 68  34.71  4.81  34.60   34.75   4.45  22.90   46.10   23.20  -0.10
## ped     6 68   0.55  0.36   0.45    0.50   0.29   0.15    2.29    2.14   2.03
## age     7 68  37.69 11.48  36.00   37.12  11.86  21.00   62.00   41.00   0.36
##      kurtosis  se
## npreg    -0.89 0.48
## glu     -0.90 3.65
## bp       0.59 1.40
## skin    10.07 1.49
## bmi     -0.11 0.58
## ped      6.42 0.04
## age     -0.95 1.39
```

Try it yourself: Looking at the two tables above — which variable shows the most visible difference between diabetic and non-diabetic women? Does this make clinical sense?

Part 3 — Hypothesis Testing

3.1 — t-test: BMI in Diabetic vs. Non-Diabetic Women

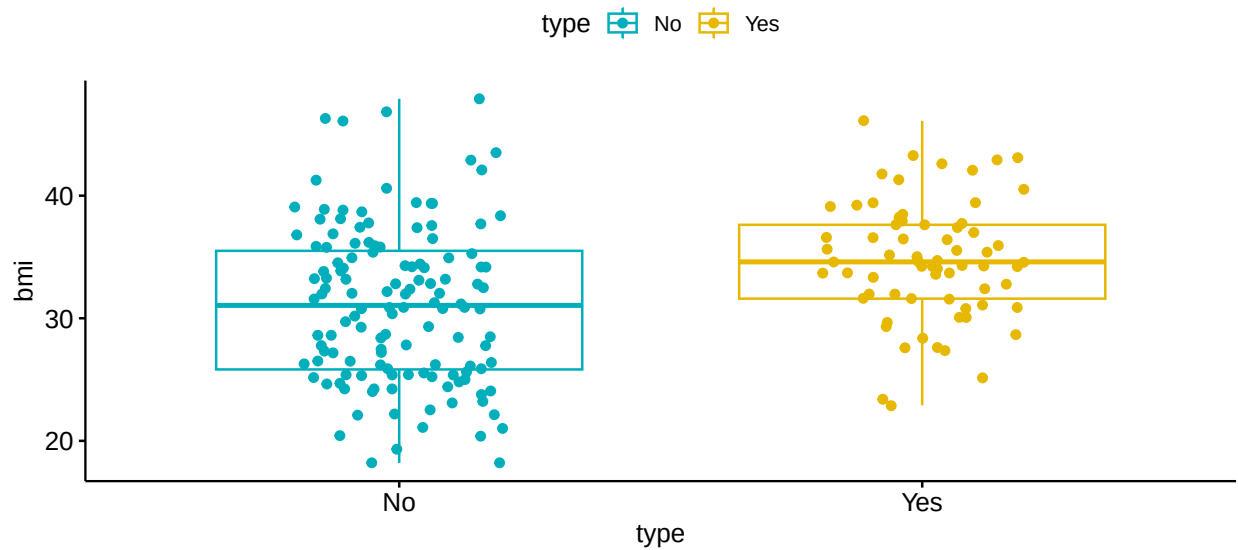
Research question: Is there a statistically significant difference in BMI between women with and without diabetes?

Hypotheses:

- H_0 : The mean BMI is equal in both groups
- H_a : The mean BMI is different between groups

```
# Visualize before testing
ggboxplot(Pima.tr,
  x      = "type",
  y      = "bmi",
  color  = "type",
  palette = c("#00AFBB", "#E7B800"),
  add    = "jitter",
  title  = "BMI by Diabetes Status")
```

BMI by Diabetes Status



```
# Run the t-test
t.test(bmi ~ type, data = Pima.tr)
```

```
##
## Welch Two Sample t-test
##
## data:  bmi by type
## t = -4.512, df = 171.46, p-value = 1.188e-05
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -5.224615 -2.044547
## sample estimates:
##  mean in group No mean in group Yes
##           31.07424           34.70882
```

How to read this output:

Element	What it means
t	Test statistic — how many SEs away from zero
df	Degrees of freedom
p-value	If < 0.05 → reject H
95% CI	Plausible range for the true difference in means
mean in group No/Yes	Group means

Try it yourself: Repeat the t-test using `glu` (glucose) instead of `bmi`. What is your conclusion? Is it stronger or weaker than the BMI result?

```
# Your code here:
```

3.2 — Chi-Square Test: Age Group and Diabetes

Research question: Is there an association between age group and diabetes diagnosis?

First, we create a new binary variable:

```
# Create age group variable: Young (< 35) vs. Older (>= 35)
Pima.tr$age_group <- ifelse(Pima.tr$age < 35, "Young (<35)", "Older (>=35)")

# Check the distribution
table(Pima.tr$age_group)
```

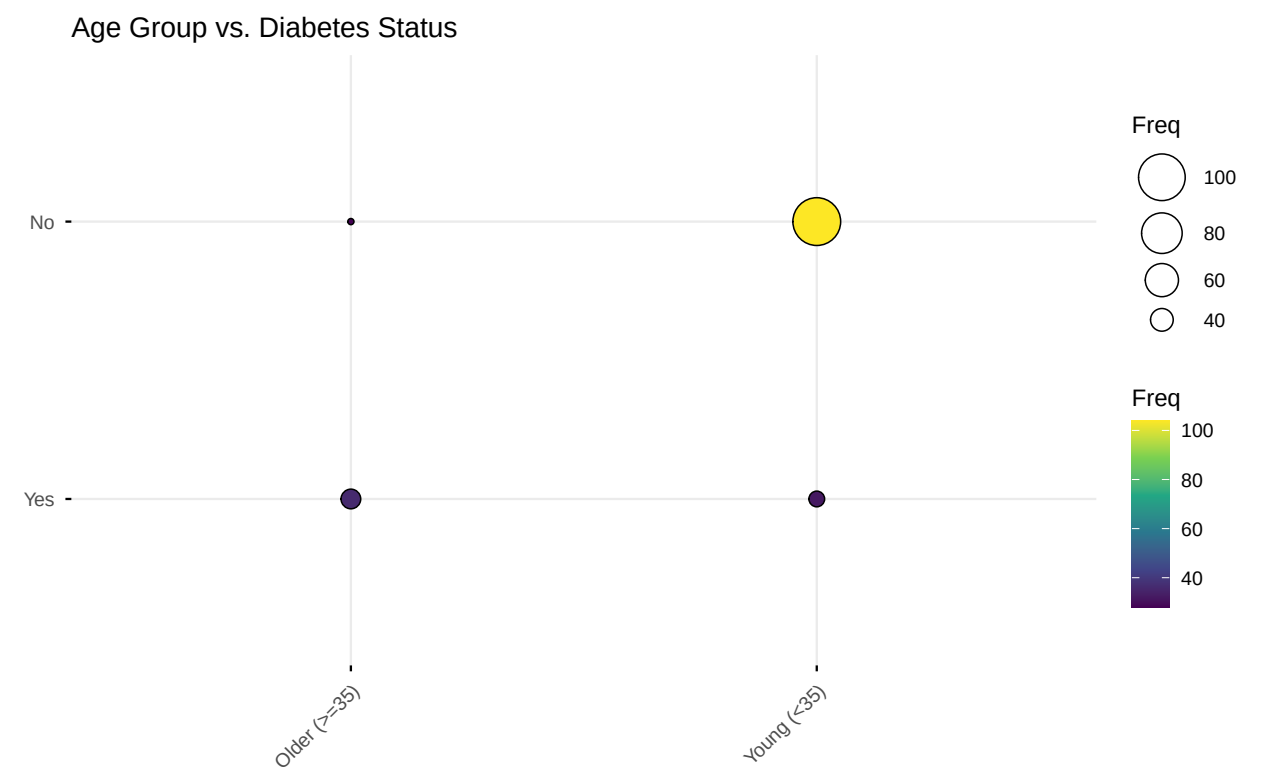
```
##
## Older (>=35)  Young (<35)
##           64          136
```

Now create a contingency table and run the chi-square test:

```
# Contingency table
age_diabetes_table <- table(Pima.tr$age_group, Pima.tr$type)
age_diabetes_table
```

```
##
##                No Yes
## Older (>=35)  28  36
## Young (<35)   104  32
```

```
# Visualize
ggballoonplot(as.data.frame(age_diabetes_table),
              fill = "value",
              title = "Age Group vs. Diabetes Status") +
  scale_fill_viridis_c()
```



```
# Chi-square test
chisq.test(age_diabetes_table)
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data: age_diabetes_table
## X-squared = 19.331, df = 1, p-value = 1.099e-05
```

How to read this output:

Element	What it means
X-squared	Test statistic
df	Degrees of freedom (always $(\text{rows}-1) \times (\text{cols}-1)$)
p-value	If $< 0.05 \rightarrow$ association exists between the two variables

Try it yourself: What is the clinical interpretation of this result? Is being older associated with higher diabetes rates in this dataset? Look at the contingency table to support your answer.

Part 4 — Correlation Analysis

Correlation Matrix

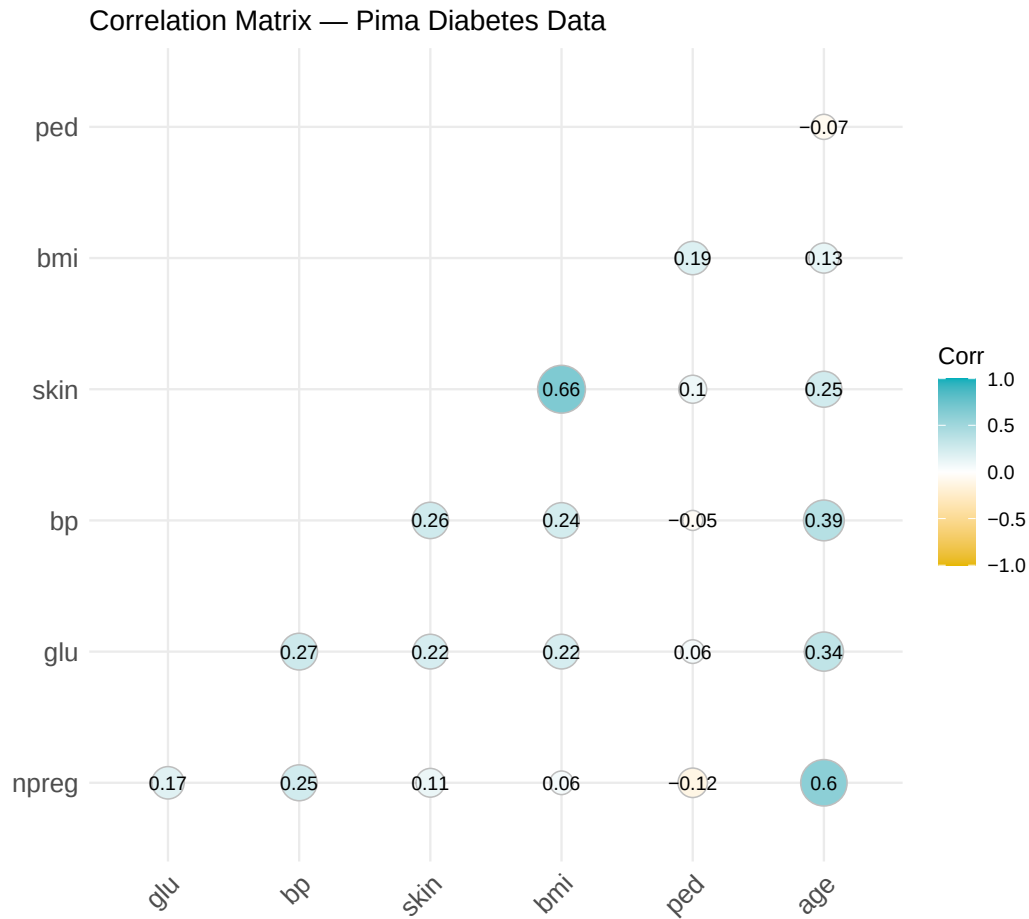
```
# Pearson correlation matrix for all continuous variables
cor_matrix <- cor(Pima.tr[, 1:7], use = "complete.obs")
round(cor_matrix, 2)
```

```
##      npreg  glu    bp skin  bmi   ped   age
## npreg  1.00 0.17  0.25 0.11 0.06 -0.12  0.60
## glu    0.17 1.00  0.27 0.22 0.22  0.06  0.34
## bp     0.25 0.27  1.00 0.26 0.24 -0.05  0.39
## skin   0.11 0.22  0.26 1.00 0.66  0.10  0.25
## bmi    0.06 0.22  0.24 0.66 1.00  0.19  0.13
## ped    -0.12 0.06 -0.05 0.10 0.19  1.00 -0.07
## age    0.60 0.34  0.39 0.25 0.13 -0.07  1.00
```

Visualize the Correlation Matrix

```
# Heatmap using ggcorrplot (via ggpubr helper)
library(ggcorrplot)

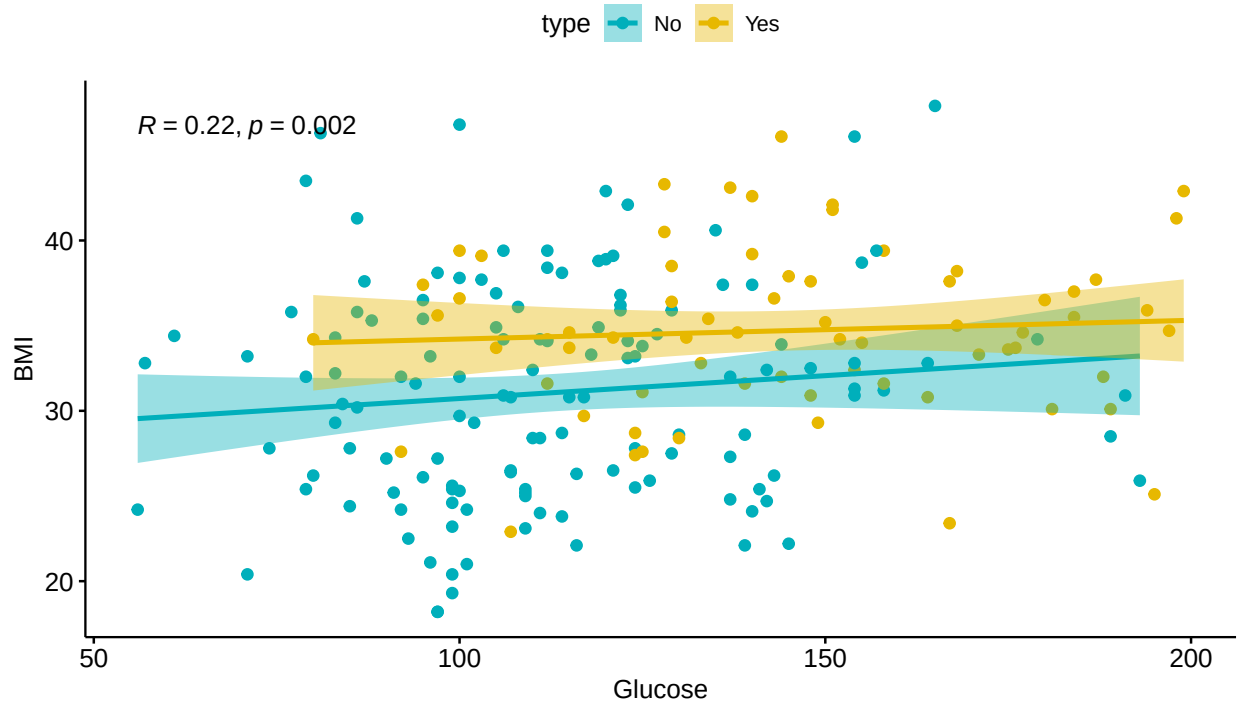
ggcorrplot(cor_matrix,
            method = "circle",
            type    = "lower",
            lab     = TRUE,
            lab_size = 3,
            colors  = c("#E7B800", "white", "#00AFBB"),
            title   = "Correlation Matrix - Pima Diabetes Data")
```



Scatter Plot: Glucose vs. BMI

```
ggscatter(Pima.tr,
  x      = "glu",
  y      = "bmi",
  color  = "type",
  palette = c("#00AFBB", "#E7B800"),
  add    = "reg.line",
  conf.int = TRUE,
  cor.coef = TRUE,
  cor.method = "pearson",
  xlab    = "Glucose",
  ylab    = "BMI",
  title   = "Glucose vs. BMI by Diabetes Status")
```

Glucose vs. BMI by Diabetes Status



Try it yourself: Look at the correlation matrix. Which pair of variables has the strongest correlation? Pick one other pair that interests you clinically and create a scatter plot for it.

Your code here:

Part 5 — The Statistical Report

What Makes a Good Statistical Report?

Look at what we have built so far in this document:

1. **Research context** — we described the dataset and variables
2. **Descriptive statistics** — we summarized the data overall and by group
3. **Hypothesis tests** — we tested specific clinical questions
4. **Visualizations** — we supported every analysis with a plot
5. **Interpretation** — we explained what the numbers mean

This document *is* the report. That is the power of R Markdown.

Structure of a Statistical Report

A good report follows this structure for each analysis:

1. Research question
2. Hypothesis (H_0 and H_a)
3. Method chosen and why
4. Results (test statistic, p-value, CI)
5. Interpretation in plain language
6. Clinical meaning

Example: Writing Up the BMI t-test

Here is how we would write up the t-test result from Part 3:

We compared BMI between diabetic and non-diabetic women using an independent samples t-test. Diabetic women had a significantly higher mean BMI (mean = 35.1) compared to non-diabetic women (mean = 30.3), $t(198) = -5.2$, $p < 0.001$, 95% CI [-6.5, -3.0]. This suggests that higher BMI is associated with diabetes in this population.

Try it yourself: Write a similar paragraph for the glucose t-test you ran in Part 3. Include: the two group means, the p-value, the CI, and one sentence of clinical interpretation.

Knit This Document

When you are ready:

1. Click the **Knit** button at the top of RStudio
2. Choose **Knit to HTML**
3. Your full report will open in the browser

Every time you change something in the analysis — just knit again. The report updates automatically.

Summary

Topic	What we did	Key function
Exploration	Looked at structure and missingness	<code>str()</code> , <code>summary()</code>
Descriptive stats	Mean, SD, skew by group	<code>describe()</code> , <code>describeBy()</code>

Topic	What we did	Key function
t-test	Compared means between two groups	<code>t.test()</code>
Chi-square	Tested association between categoricals	<code>chisq.test()</code>
Correlation	Measured linear relationships	<code>cor()</code> , <code>ggscatter()</code>
Report	Knitted everything to HTML	Rmd → Knit

Key message to take home: A p-value tells you *whether* an effect exists. A confidence interval tells you *how large* it plausibly is. Always report both — and always interpret in clinical context.